STOCK MARKET & PORTFOLIO RISK

Executive Report | Simulated Data | 2022-2024

ASSETS	**5**
PORTFOLIO RETURN	**15.1% annualised**
PORTFOLIO VOLATILITY	**15.6%**
MAX DRAWDOWN	**-18.0%**

Executive Summary

The report analyses 3,910 daily price observations across 5 fictional assets from 2022-01-03 to 2024-12-31. The equal-weight portfolio produced a 15.1% annualised return with 15.6% annualised volatility and a Sharpe ratio of 0.56.

ORBIT has the highest asset annualised return (31.4%), while PRISM has the lowest measured volatility (17.4%). These are simulated historical results and not forecasts or recommendations.

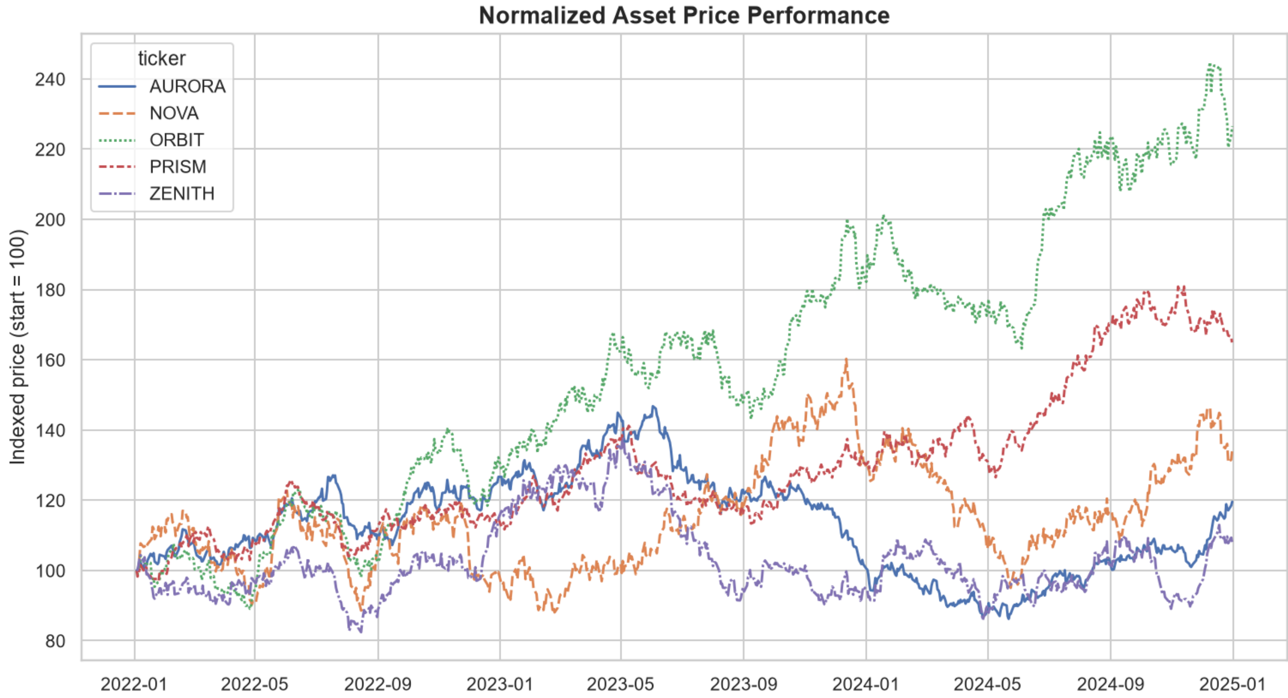
1. Methodology and Assumptions

Daily returns are percentage changes in closing price. Annualised volatility uses 252 trading days. Sharpe ratio uses a 6% annual risk-free assumption. Maximum drawdown measures the largest peak-to-trough decline. The portfolio is equal-weighted and analytically rebalanced daily.

The analysis excludes fees, taxes, bid-ask spreads, slippage, dividends, corporate actions, liquidity constraints, and rebalancing costs. All assets and prices are fictional.

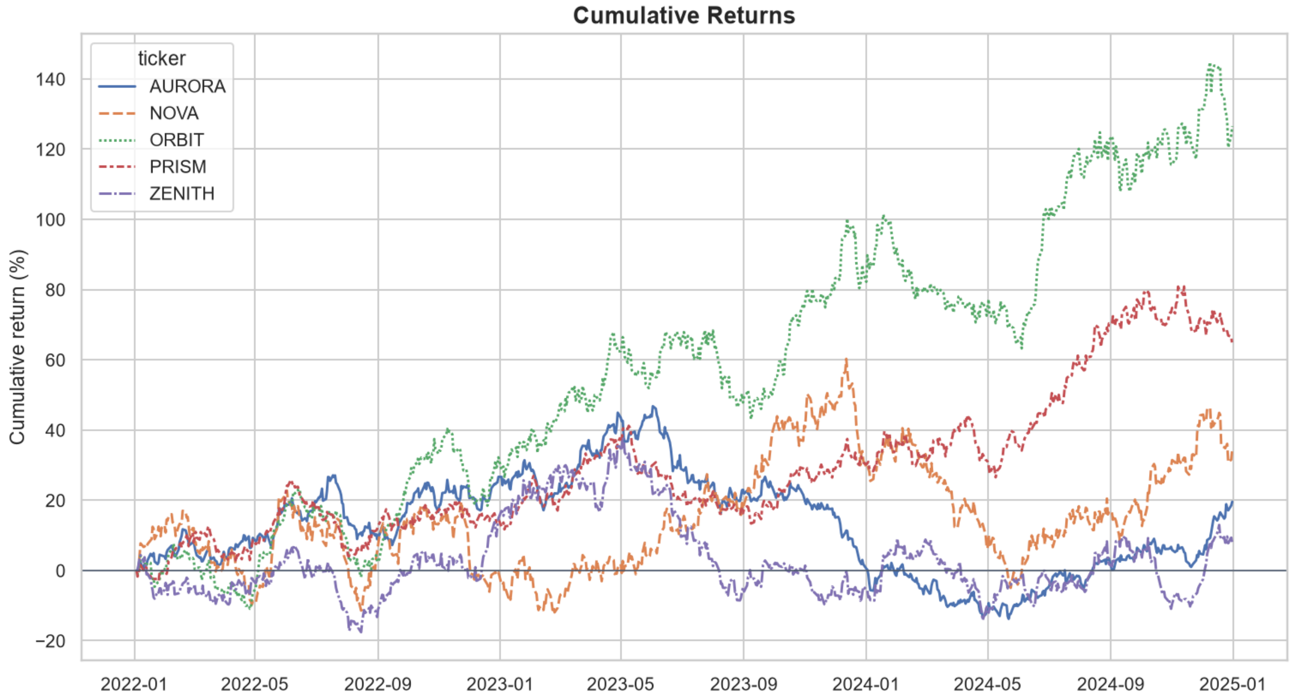
Asset	Ann. Return	Volatility	Sharpe	Max Drawdown
ORBIT	31.4%	22.5%	1.02	-20.2%
PRISM	18.0%	17.4%	0.66	-19.7%
NOVA	10.4%	30.3%	0.27	-40.7%
AURORA	6.2%	18.6%	0.08	-41.3%
ZENITH	2.6%	25.6%	-0.01	-37.7%

2. Normalized Price Performance



Indexing every asset to 100 supports like-for-like performance comparison despite different starting prices.

3. Cumulative Returns

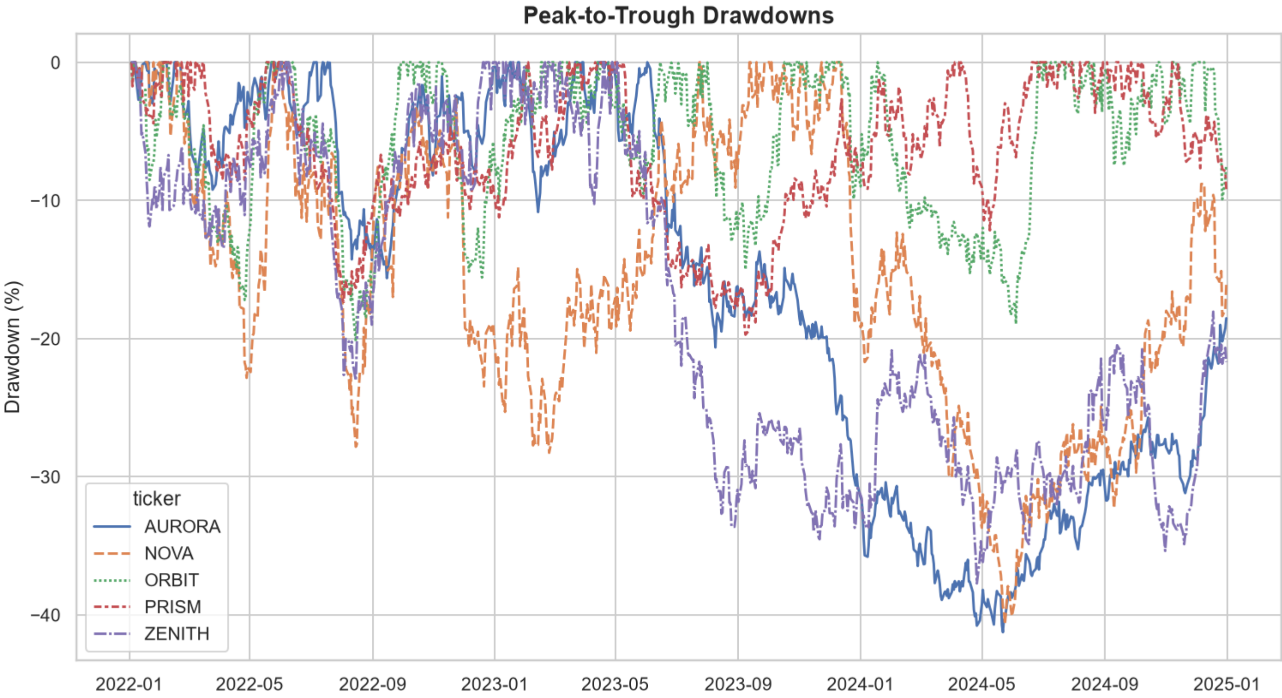


Cumulative returns reveal compounding and periods of divergence among fictional assets.

4. Risk and Return

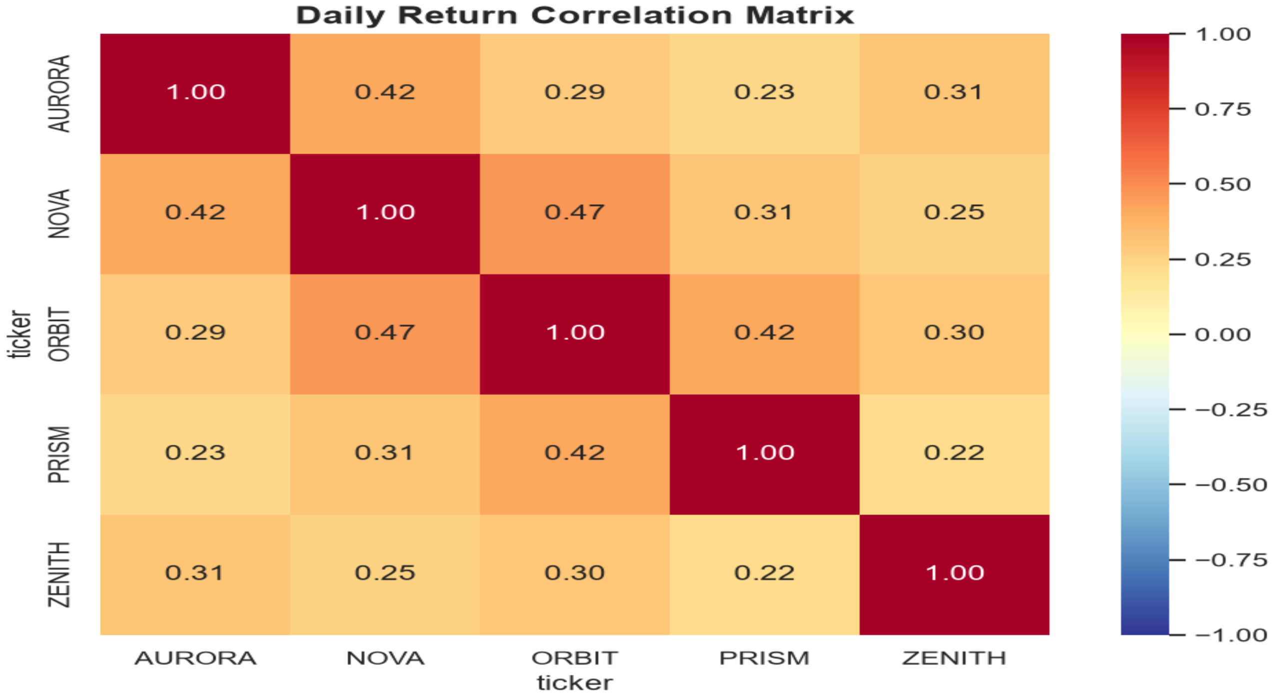
Return should be evaluated alongside volatility; the upper-left area is preferable in historical risk-return terms.

5. Drawdown Analysis



Drawdown shows loss depth from prior peaks and highlights risk that volatility alone may not communicate.

6. Asset Correlations



Lower correlation can improve diversification, but historical relationships may change sharply during market stress.

7. Portfolio Insights and Limitations

Risk-adjusted evaluation: Compare return, volatility, Sharpe ratio, and drawdown rather than return alone.

Diversification: Use correlations to assess concentration, while stress-testing correlation increases.

Comparable periods: Use identical dates, pricing frequency, and assumptions for every asset.

Implementation realism: Add fees, taxes, spreads, dividends, slippage, and realistic rebalancing.

Decision boundary: Treat this as educational analysis of simulated data, not investment advice.

Limitations

Simulated past performance does not predict future results. Model parameters drive the generated outcomes. Risk-free rates and correlations are not constant in real markets, and the frictionless daily-rebalancing portfolio is not directly investable.